Lectures on Monte Carlo Theory

Chapter 5: Variance Reduction Techniques

Based on Lorek & Rolski

Chapter 5 Outline

Motivation: Why Reduce Variance? I

The crude Monte Carlo (CMC) estimator computes $I = \mathbb{E}[Y]$ by averaging R independent copies of Y:

$$\hat{Y}_R^{\mathrm{CMC}} = \frac{1}{R}\sum_{j=1}^{R} Y_j,$$

where $Y_1, \ldots, Y_R$ are i.i.d. replications of Y, R (capital R) is the number of replications, and $\hat{Y}_R^{\mathrm{CMC}}$ (Y-hat, CMC) is the estimate of $I = \mathbb{E}[Y]$ (expected value, i.e., average, of Y).
The variance of this estimator is $\mathrm{Var}(\hat{Y}_R^{\mathrm{CMC}}) = \mathrm{Var}(Y)/R$, meaning the variance (spread of estimates around the true value) shrinks as we use more replications, but at a slow rate.

The Fundamental Problem: Diminishing Returns

The half-width of a 95% confidence interval is $b \approx 1.96\,\sigma_Y/\sqrt{R}$, where σ_Y (sigma, meaning standard deviation of Y) is the square root of $\mathrm{Var}(Y)$. To halve the error b, we need *four times* as many samples. Reducing $\mathrm{Var}(Y)$ by a factor k is equivalent to running k times more simulations – without the extra cost.

Motivation: Why Reduce Variance? II

The goal of variance reduction is to construct a new estimator $\hat{Y}_R^{\text{new}}$ of I satisfying two requirements:

1. **Consistency:** $\hat{Y}_R^{\text{new}} \to I$ with probability 1 as $R \to \infty$ (it converges to the right answer).
2. **Central Limit Theorem (CLT):** $\sqrt{R}(\hat{Y}_R^{\text{new}} - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, (\varsigma^{\text{new}})^2)$ where ς (varsigma, a variant of sigma) is the asymptotic standard deviation, and $\mathcal{D}$ (D with arrow) means "converges in distribution."

The quantity $(\varsigma^{\text{new}})^2$ is called the *time-average variance constant* (TAVC) or *asymptotic variance*, defined as $(\varsigma^{\text{new}})^2 = \lim_{R\to\infty} R\,\mathrm{Var}(\hat{Y}_R^{\text{new}})$. A smaller TAVC means a tighter confidence interval for the same computational budget.

Motivation: Why Reduce Variance? III

Methods Covered in This Chapter

1. Stratified sampling (str)
2. Antithetic variates (anti)
3. Common random numbers (CRN)
4. Ratios of expectations (RoE)
5. Control variates (CV)
6. Conditional Monte Carlo (cond)
7. Importance sampling (IS)
8. Cross-entropy method (CE)

5.1.1 Stratified Sampling – Idea and Setup I

Stratified sampling is based on a simple but powerful idea: instead of sampling Y from its full distribution at once, we divide the sample space into non-overlapping *strata* (regions) and sample separately from each one. This prevents the random allocation of effort across regions and instead guarantees a prescribed number of samples in each region.

Setup (Definition 5.1)

Partition the sample space Ω (omega, the entire probability space) into m disjoint strata $S^1, \ldots, S^m$ with $\bigcup_{j=1}^m S^j = \Omega$ and $\mathbb{P}(Y \in S^j) = p_j > 0$. The probabilities p_j (p sub j) satisfy $\sum_{j=1}^m p_j = 1$.
Define:

- $I^j = \mathbb{E}[Y \mid Y \in S^j]$: the conditional expectation of Y within stratum j (typically unknown).
- $\sigma_j^2 = \mathrm{Var}(Y^j) = \mathrm{Var}(Y \mid Y \in S^j)$: the variance within stratum j (sigma squared sub j).

Then $I = \mathbb{E}[Y] = p_1 I^1 + \cdots + p_m I^m$ (law of total expectation).

5.1.1 Stratified Sampling – Idea and Setup II

Law of Total Variance (Proposition 5.1.1)

For any two random variables Y and J (where J randomly selects the stratum with $\mathbb{P}(J = j) = p_j$):

$$\mathrm{Var}(Y) = \underbrace{\mathbb{E}[\mathrm{Var}(Y|J)]}_{\text{within-stratum}} + \underbrace{\mathrm{Var}(\mathbb{E}[Y|J])}_{\text{between-stratum}}.$$

Expanding these two terms (Corollary 5.1.2):

$$\mathrm{Var}(Y) = \sum_{j=1}^{m} p_j \sigma_j^2 + \sum_{j=1}^{m} p_j (I^j - I)^2.$$

The first sum is the average within-stratum variance; the second is the variance between stratum means.

The key insight: the stratified estimator eliminates the between-stratum term $\sum_j p_j (I^j - I)^2$, which can be large if stratum means differ greatly.

5.1.1 Stratified Sampling – Idea and Setup III

The Stratified Estimator

Let $Y_1^j, \ldots, Y_{R_j}^j$ be i.i.d. samples from stratum j (that is, $Y_i^j \stackrel{d}{=} (Y \mid Y \in S^j)$), independent across strata. The within-stratum sample mean is:

$$\hat{Y}_{R_j}^j = \frac{1}{R_j} \sum_{i=1}^{R_j} Y_i^j,$$

where R_j (R sub j) is the number of replications in stratum j, with total $R = R_1 + \cdots + R_m$. The *stratified estimator* combines the stratum estimates:

$$\hat{Y}_R^{\mathrm{str}} = \sum_{j=1}^{m} p_j \hat{Y}_{R_j}^j = p_1 \hat{Y}_{R_1}^1 + \cdots + p_m \hat{Y}_{R_m}^m.$$

This is unbiased: $\mathbb{E}[\hat{Y}_R^{\mathrm{str}}] = I$, and its variance is:

$$\mathrm{Var}(\hat{Y}_R^{\mathrm{str}}) = \sum_{j=1}^{m} \frac{p_j^2 \sigma_j^2}{R_j}.$$

5.1.1 Stratified Sampling – Worked Example I

Example 5.1.8 – Estimating π via Stratified Sampling

We estimate π using the quarter-circle area. Recall that for $U \sim \mathcal{U}[0,1]$ (U uniformly distributed on $[0,1]$):

$$Y = 4\sqrt{1-U^2}, \qquad \mathbb{E}[Y] = 4\int_0^1 \sqrt{1-u^2}\,du = \pi.$$

Define m equally probable strata $B^j = \left(\frac{j-1}{m}, \frac{j}{m}\right]$ for $j = 1, \ldots, m$, so $p_j = 1/m$ for all j. Within stratum j, sample $V^j = (j-1)/m + U/m$ where $U \sim \mathcal{U}[0,1]$, and compute $Y^j = 4\sqrt{1-(V^j)^2}$. The stratified estimator with proportional allocation ($R_j = R/m$) is:

$$\hat{Y}_R^{\text{str}} = \frac{1}{m}\sum_{j=1}^{m} \hat{Y}^j_{R_j} = \frac{1}{R}\sum_{j=1}^{m}\sum_{i=1}^{R/m} 4\sqrt{1-(V_i^j)^2}.$$

Numerical results from Table 5.2 (book) with $R = 10^4$:

Strata m	CMC	Str (prop, $m = 10$)	Str (prop, $m = 20$)
Half-width b	0.0177	0.00400	0.00209
Var reduction	$1\times$	$\approx 20\times$	$\approx 72\times$

Table 5.1 from the book shows the pilot-estimated standard deviations $\hat{S}'_j$ for $m = 10$ strata (with $R' = 100$ pilot simulations):

Stratum j	1	2	3	4	5	6	7	8	9	10
$\hat{S}'_j$	0.006	0.018	0.031	0.034	0.059	0.078	0.112	0.097	0.169	0.453
R_j (opt)	1	3	5	6	11	14	21	18	31	90

5.1.1 Stratified Sampling – Worked Example III

Notice that stratum 10 (the region near $U \approx 1$) receives 90 out of 200 replications, because the function $4\sqrt{1-u^2}$ is steeply curved there and has high variability.

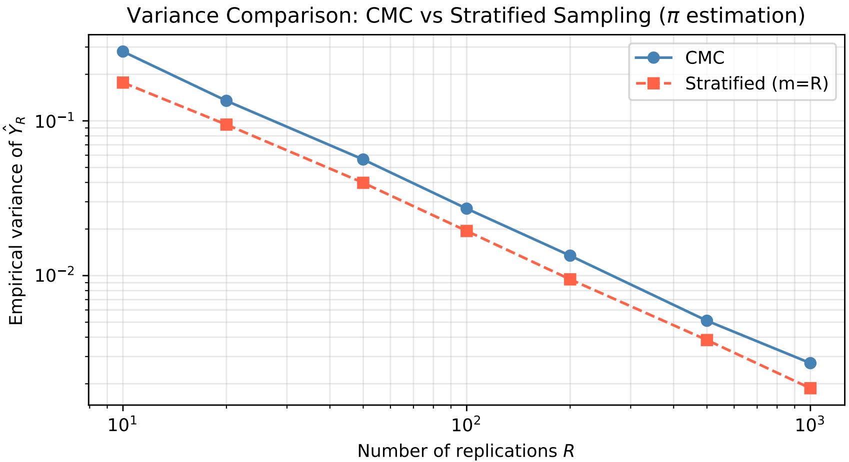
How to Read This

The optimal allocation shifts effort away from the easy strata (near $j = 1$, where Y is nearly constant at 4) and toward the hard strata (near $j = m$, where Y drops steeply to 0 and σ_j is large).

5.1.1 Stratified Sampling – Python Code

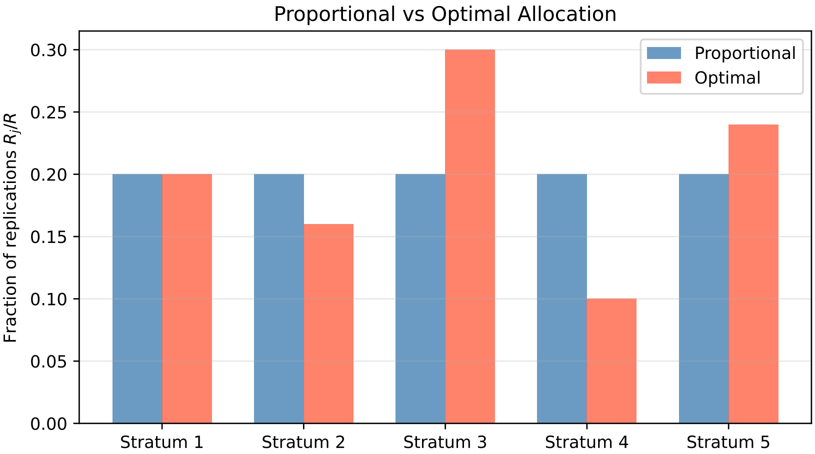
```python
import numpy as np

def stratified_pi(m, R, use_optimal=False, R_pilot=100):
    """Estimate pi using stratified sampling on U with m strata."""
    if use_optimal:
        # Pilot phase: estimate sigma_j
        R_j_pilot = R_pilot // m
        sigma_hat = []
        for j in range(m):
            V = (j + np.random.uniform(0, 1, R_j_pilot)) / m
            Yj = 4 * np.sqrt(1 - V**2)
            sigma_hat.append(np.std(Yj, ddof=1))
        sigma_hat = np.array(sigma_hat)
        # Optimal allocation
        R_j = np.maximum(1, np.round(sigma_hat / sigma_hat.sum() * R).astype(int))
    else:
        R_j = np.full(m, R // m)  # proportional (equal p_j = 1/m)

    estimates = []
    for j in range(m):
        V = (j + np.random.uniform(0, 1, R_j[j])) / m
        Yj = 4 * np.sqrt(1 - V**2)
        estimates.append(np.mean(Yj))
    return np.mean(estimates)  # equal p_j = 1/m

# Comparison: CMC vs stratified
np.random.seed(42)
```

5.1.1 Stratified Sampling – Variance Comparison



Stratified sampling reduces variance significantly – the gap widens as m (number of strata) increases. Optimal allocation beats proportional, which beats CMC. Both converge to π as $R \to \infty$, but with different rates.

5.1.1 Optimal vs. Proportional Allocation



Optimal allocation assigns more samples to strata with larger σ_j (within-stratum standard deviation), regardless of stratum probability p_j. For $Y = 4\sqrt{1-U^2}$, the rightmost stratum (steep curve near $U = 1$) has by far the highest variability and receives the most replications.

5.1.2 Antithetic Variates I

Antithetic variates reduce variance by generating pairs of estimates that are *negatively correlated*: when one overestimates I, the other tends to underestimate it, so that their average cancels out the errors.

5.1.2 Antithetic Variates II

The Antithetic Estimator

Instead of generating R independent copies of Y, generate $R/2$ pairs $(Y_1^*, Y_1^{*\prime}), \ldots, (Y_{R/2}^*, Y_{R/2}^{*}{}')$ such that:

- $Y_i^* \overset{d}{=} Y$ and $Y_i^{*\prime} \overset{d}{=} Y$ (both have the correct marginal distribution),
- Pairs (Y_{2i-1}^*, Y_{2i}^*) are i.i.d. (independent across pairs).

The antithetic estimator is:

$$\hat{Y}_R^* = \frac{1}{R}\sum_{j=1}^{R} Y_j^* = \frac{1}{R/2}\sum_{i=1}^{R/2} \frac{Y_{2i-1}^* + Y_{2i}^*}{2}.$$

Its variance compared to CMC (using (5.1.15) from the book):

$$\mathrm{Var}(\hat{Y}_R^*) = \mathrm{Var}(\hat{Y}_R^{\mathrm{CMC}}) \cdot \big(1 + \mathrm{Corr}(Y_1^*, Y_2^*)\big).$$

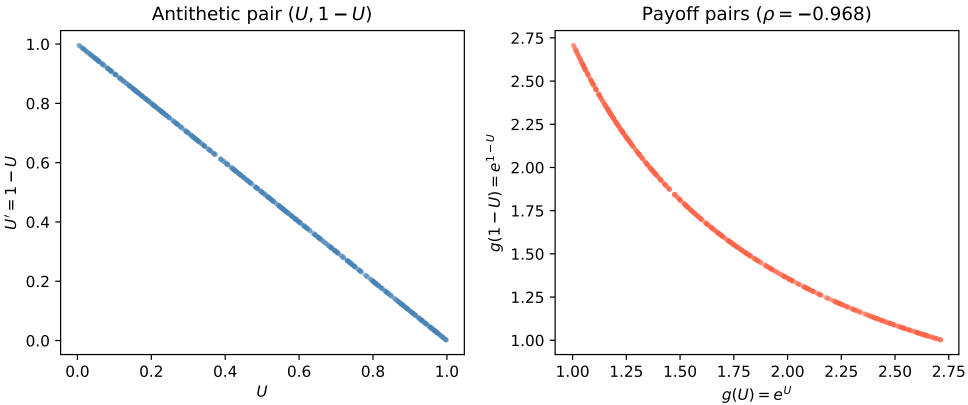
Here Corr (correlation, ranging from -1 to $+1$) between the two members of a pair. If $\mathrm{Corr}(Y_1^*, Y_2^*) < 0$, variance decreases.

Standard Construction: Complement in [0, 1]

5.1.2 Antithetic Variates – Python Code

```python
import numpy as np

def antithetic_exp(R):
    """Estimate E[e^U], U~U[0,1], using antithetic variates."""
    U  = np.random.uniform(0, 1, R // 2)
    Y1 = np.exp(U)              # original
    Y2 = np.exp(1 - U)          # antithetic (complement)
    return np.mean((Y1 + Y2) / 2)

def antithetic_pi(R):
    """Estimate pi using antithetic variates: Y = 4*sqrt(1-U^2)."""
    U  = np.random.uniform(0, 1, R // 2)
    Y1 = 4 * np.sqrt(1 - U**2)
    Y2 = 4 * np.sqrt(1 - (1-U)**2)    # Y(1-U) is antithetic
    return np.mean((Y1 + Y2) / 2)

np.random.seed(42)
R = 100000
# E[e^U] = e - 1 ~ 1.71828
print(f"Antithetic E[e^U] = {antithetic_exp(R):.6f}  (true: {np.e-1:.6f})")
print(f"Antithetic pi     = {antithetic_pi(R):.6f}  (true: {np.pi:.6f})")

# Variance comparison: run 2000 repetitions of size 1000
cmc_vars  = np.var([np.mean(4*np.sqrt(1-np.random.uniform(0,1,1000)**2))
                    for _ in range(2000)])
anti_vars = np.var([antithetic_pi(1000) for _ in range(2000)])
print(f"CMC variance:       {cmc_vars:.6f}")
```

5.1.2 Antithetic Variates – Illustration



Left: The pair $(U, 1-U)$ is perfectly negatively correlated. Right: The corresponding payoffs $k(U)$ and $k(1-U)$ also show strong negative correlation, so their average has much lower variance than either one alone.

5.1.3 Common Random Numbers (CRN) I

Common random numbers (CRN) is a technique for *comparing two systems*: instead of estimating a single $\mathbb{E}[Y]$, we want to estimate the difference $\Delta = \mathbb{E}[Y_1] - \mathbb{E}[Y_2]$ between outputs of two system configurations. By using the *same* random numbers to drive both simulations, we induce positive correlation between Y_1 and Y_2, which reduces the variance of the difference estimator.

5.1.3 Common Random Numbers (CRN) II

CRN Estimator

Use the same random stream U to simulate both systems. Generate R pairs:

$$\hat{\Delta}_R^{\text{CRN}} = \frac{1}{R}\sum_{j=1}^{R}(Y_{1,j} - Y_{2,j}),$$

where $(Y_{1,j}, Y_{2,j})$ are the outputs of systems 1 and 2 using the same random input U_j. The variance is:

$$\text{Var}(\hat{\Delta}_R^{\text{CRN}}) = \frac{\text{Var}(Y_1) + \text{Var}(Y_2) - 2\text{Cov}(Y_1, Y_2)}{R}.$$

CRN reduces variance when $\text{Cov}(Y_1, Y_2) > 0$, which is the common case when both systems respond monotonically to the same random input.

5.1.3 Common Random Numbers (CRN) III

Why CRN induces positive covariance: If both $Y_1 = k_1(U)$ and $Y_2 = k_2(U)$ are non-decreasing functions of the same input U, then Theorem 5.1.13 gives $\mathrm{Cov}(k_1(U), k_2(U)) \geq 0$.

Example – Scheduling Problem (Example 5.1.20)

Consider two scheduling policies for a queue (FCFS vs. LCFS ordering). We want to estimate the difference in expected throughput Δ. Using CRN:

- The same sequence of service times and inter-arrival times is fed to both policies in each replication.
- When service times happen to be long (bad run), both policies suffer similarly, so $Y_1 - Y_2$ is stable.
- Result from book: for $R = 1000$, variance reduced by factor $\approx 22.5\times$ compared to independent replications.

5.1.3 Common Random Numbers (CRN) IV

Synchronisation Warning

When systems have different structures (different number of events, different branching), naive use of CRN may fail to induce positive correlation. Careful mapping of the random number stream to the same logical variable in both systems (called *synchronisation*) is required.

5.1.4 Ratios of Expectations I

Sometimes the quantity of interest is itself a ratio of two expectations: $I = I^{(1)}/I^{(2)}$ where $I^{(1)} = \mathbb{E}[Y^{(1)}]$ and $I^{(2)} = \mathbb{E}[Y^{(2)}]$. This arises, for example, in conditional expectations and self-normalised importance sampling.

Ratio Estimator

Generate R i.i.d. pairs $(Y_1^{(1)}, Y_1^{(2)}), \ldots, (Y_R^{(1)}, Y_R^{(2)})$ and define:

$$\hat{Y}_R^{\mathrm{RoE}} = \frac{\sum_{i=1}^R Y_i^{(1)}}{\sum_{i=1}^R Y_i^{(2)}} = \frac{\hat{Y}_R^{(1)}}{\hat{Y}_R^{(2)}},$$

where $\hat{Y}_R^{(1)}$ and $\hat{Y}_R^{(2)}$ are the sample means. This estimator is *biased* (the ratio of sample means is not the same as the sample mean of ratios), but it is *consistent* (converges to I as $R \to \infty$).

5.1.4 Ratios of Expectations II

Asymptotic Variance via the Delta Method

The *delta method* (also called the propagation of uncertainty formula) approximates the variance of a function of sample means. For the ratio estimator:

$$(\varsigma^{\mathrm{RoE}})^2 = \frac{I^{(1)}}{I^{(2)}} \left[\frac{\Sigma_{\mathbf{Y}}(1,1)}{(I^{(1)})^2} - \frac{2\,\Sigma_{\mathbf{Y}}(1,2)}{I^{(1)} I^{(2)}} + \frac{\Sigma_{\mathbf{Y}}(2,2)}{(I^{(2)})^2} \right]^{1/2},$$

where $\Sigma_{\mathbf{Y}}(i,j) = \mathrm{Cov}(Y^{(i)}, Y^{(j)})$. By choosing the joint distribution of $(Y^{(1)}, Y^{(2)})$ to induce positive covariance $\Sigma_{\mathbf{Y}}(1,2)$, we can reduce the variance of the ratio estimator. This is the connection to common random numbers (CRN).

Application: Self-Normalised Importance Sampling

Self-normalised IS (see Section 5.1.7) uses a ratio estimator when the normalisation constant of a distribution is unknown. The ratio estimator is consistent and the delta method provides the confidence interval.

5.1.5 Control Variates I

A control variate is a random variable X that is *correlated* with the output Y and has a *known mean* $\mu_X = \mathbb{E}[X]$. The idea is to correct the CMC estimate $\hat{Y}_R$ by subtracting the observed deviation of X from its known mean.

The Control Variate Estimator

Given primary output Y with $I = \mathbb{E}[Y]$ (unknown), control variate X with $\mu_X = \mathbb{E}[X]$ (known), and scalar coefficient c (c for control):

$$\hat{Y}_R^{\mathrm{cv}} = \hat{Y}_R + c\,(\hat{X}_R - \mu_X),$$

where $\hat{Y}_R = \frac{1}{R}\sum_i Y_i$ and $\hat{X}_R = \frac{1}{R}\sum_i X_i$ are sample means from the same R replications. This estimator is *unbiased* because $\mathbb{E}[\hat{X}_R - \mu_X] = 0$. Its variance is:

$$\mathrm{Var}(\hat{Y}_R^{\mathrm{cv}}) = \frac{1}{R}\big[\mathrm{Var}(Y) + c^2\,\mathrm{Var}(X) + 2c\,\mathrm{Cov}(X,Y)\big].$$

5.1.5 Control Variates II

Optimal Coefficient c^* and Resulting Variance

Minimising the variance over c gives:

$$c^* = -\frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}.$$

Here $\mathrm{Cov}(X,Y)$ (covariance of X and Y, measuring how they move together) divided by $\mathrm{Var}(X)$ (variance of X) gives the regression slope of Y on X. The optimal variance is:

$$\mathrm{Var}(\hat{Y}_R^{\mathrm{cv},*}) = \frac{\mathrm{Var}(Y)}{R}(1-\rho^2),$$

where ρ (rho) $= \mathrm{Corr}(X,Y)$ is the correlation between X and Y. The *variance reduction factor* is $1/(1-\rho^2)$, which is always ≥ 1.

5.1.5 Control Variates III

How to Read This

If $\rho^2 = 0.9$ (strongly correlated), the variance is reduced by factor $1/(1-0.9) = 10$. If $\rho = 0$ (uncorrelated), no reduction occurs. The control variate works by "explaining away" the variability in Y that is linearly predictable from X.

5.1.5 Control Variates IV

Estimated Coefficient (Pilot Simulation)

In practice, $c^* = -\mathrm{Cov}(X, Y)/\mathrm{Var}(X)$ is unknown. Run R' pilot replications to estimate:

$$\hat{c} = -\frac{\hat{S}^2_{XY}}{\hat{S}^2_X},$$

where $\hat{S}^2_{XY} = \frac{1}{R'-1}\sum_{i=1}^{R'}(X_i - \bar{X})(Y_i - \bar{Y})$ is the sample covariance and $\hat{S}^2_X$ is the sample variance of X. Then run R main replications and apply: $\hat{Y}^{\mathrm{cv}}_R = \hat{Y}_R + \hat{c}(\hat{X}_R - \mu_X)$.

Example 5.1.24 – π with Control Variates

Estimate $I = \pi$ using $Y = 4\sqrt{1-U^2}$, $U \sim \mathcal{U}[0,1]$.

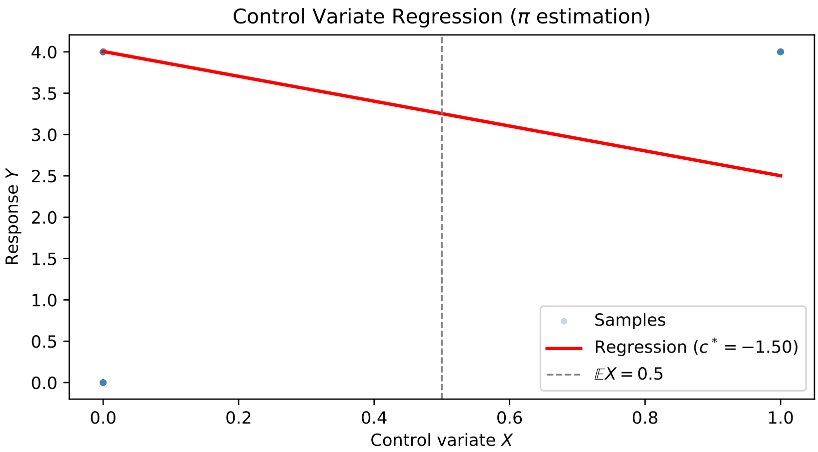
- **Control variate 1:** $X = 1 - U$ with $\mu_X = \mathbb{E}[1-U] = 1/2$.
  $\rho \approx 0.528$, variance reduction $\approx 1.4\times$.
- **Control variate 2:** $X = \mathbf{1}(U + (1-U) > \sqrt{2})$, μ_X known.
  $1 - \rho^2 = 0.242$, variance reduction $\approx 4\times$.
- **Control variate 3:** degree-4 polynomial in U tuned to Y.
  $1 - \rho^2 = 0.0000860$, variance reduction $\approx 11{,}630\times$!

The lesson: a cleverly chosen control variate can achieve enormous reductions. The more X "explains" the variability of Y (high ρ^2), the better.

5.1.5 Control Variates – Python Code

```python
import numpy as np

def control_variate_pi(R, R_pilot=1000):
    """Estimate pi using control variate X = 1-U, E[X] = 0.5."""
    # Pilot: estimate c* = -Cov(X,Y)/Var(X)
    U_p = np.random.uniform(0, 1, R_pilot)
    Y_p = 4 * np.sqrt(1 - U_p**2)    # primary output
    X_p = 1 - U_p                     # control variate (E[X] = 0.5)
    cov_mat = np.cov(X_p, Y_p, ddof=1)
    c_hat = -cov_mat[0, 1] / cov_mat[0, 0]
    rho   = np.corrcoef(X_p, Y_p)[0, 1]
    print(f"  Estimated c* = {c_hat:.4f},  rho = {rho:.4f}")
    print(f"  Theoretical reduction = {1/(1-rho**2):.2f}x")

    # Main simulation
    U = np.random.uniform(0, 1, R)
    Y = 4 * np.sqrt(1 - U**2)
    X = 1 - U
    Y_cv = Y + c_hat * (X - 0.5)     # control variate adjustment
    return np.mean(Y_cv), np.std(Y_cv) / np.sqrt(R)

np.random.seed(42)
est, se = control_variate_pi(R=100000)
print(f"CV estimate:  {est:.6f} +/- {1.96*se:.6f}")
print(f"True pi:      {np.pi:.6f}")
```

5.1.5 Control Variates – Regression View



The control variate estimator fits a regression line through the (X, Y) scatter plot. The optimal coefficient $c^* = -\mathrm{Cov}(X, Y)/\mathrm{Var}(X)$ is the negative of the regression slope. The residuals from this regression have lower variance than Y itself – that is exactly the variance reduction.

5.1.6 Conditional Monte Carlo (Rao-Blackwell) I

Conditional Monte Carlo (also called the Rao-Blackwell method) exploits the structure of a problem by computing a conditional expectation analytically instead of simulating it. If we can compute $c(x) = \mathbb{E}[Y \mid X = x]$ in closed form, then $c(X)$ is a better estimator of $I = \mathbb{E}[Y]$ than Y itself.

The Conditional Estimator

Given $I = \mathbb{E}[Y] = \mathbb{E}[c(X)]$ where $c(x) = \mathbb{E}[Y \mid X = x]$, generate $X_1, \ldots, X_R$ i.i.d. copies of X and compute:

$$\hat{Y}_R^{\text{cond}} = \frac{1}{R}\sum_{j=1}^{R} c(X_j).$$

This estimator uses the analytically computed conditional mean $c(X_j)$ instead of one noisy sample Y_j per replication.

5.1.6 Conditional Monte Carlo (Rao-Blackwell) II

Rao-Blackwell Theorem: Conditioning Always Reduces Variance

By the law of total variance:

$$\mathrm{Var}(Y) = \underbrace{\mathbb{E}[\mathrm{Var}(Y \mid X)]}_{\geq 0} + \mathrm{Var}(\mathbb{E}[Y \mid X]) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(c(X)).$$

Since $\mathbb{E}[\mathrm{Var}(Y \mid X)] \geq 0$, we have:

$$\mathrm{Var}(c(X)) \leq \mathrm{Var}(Y).$$

This is strict unless $Y = c(X)$ almost surely (i.e., Y is already a deterministic function of X).

Example 5.1.29 – π via Conditional MC

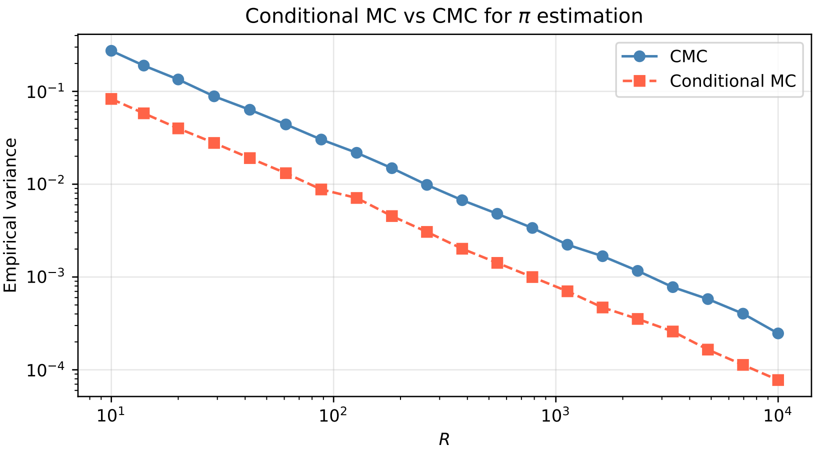
We estimate π via the quarter-circle: $Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$ where $U_1, U_2 \sim \mathcal{U}[0,1]$. Condition on $X = U_1$:

$$c(x) = \mathbb{E}[Y \mid U_1 = x] = 4\,\mathbb{P}(U_2 \leq \sqrt{1-x^2}) = 4\sqrt{1-x^2}.$$

The conditional estimator is just: $\hat{Y}_R^{\text{cond}} = \frac{4}{R}\sum_{i=1}^{R} \sqrt{1-U_i^2}$.

- $\text{Var}(\hat{Y}_R^{\text{CMC}}) = 2.697/R$
- $\text{Var}(\hat{Y}_R^{\text{cond}}) = 0.797/R$
- Reduction factor: $\approx 3.38\times$

5.1.6 Conditional MC – Variance Comparison



The conditional MC estimator uses the analytically computed $c(U_1) = 4\sqrt{1 - U_1^2}$ instead of the noisy indicator $4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$. Both converge at rate $1/R$, but the conditional estimator has a smaller constant.

5.1.7 Importance Sampling – The Change-of-Measure Idea I

Importance sampling (IS) is a powerful technique that allows us to estimate $I = \int k(x)f(x)\,dx = \mathbb{E}_f[k(X)]$ by sampling from a different (easier or better-suited) distribution $\tilde{f}$ instead of f. The key insight is that any expectation under f can be rewritten as an expectation under $\tilde{f}$ by multiplying by the ratio $f(x)/\tilde{f}(x)$.

5.1.7 Importance Sampling – The Change-of-Measure Idea II

Importance Sampling Identity

Let $\tilde{f}$ (f-tilde, the *proposal* or *importance* density) satisfy $\tilde{f}(x) > 0$ wherever $k(x)f(x) \neq 0$. Define the *likelihood ratio* (also called importance weight):

$$L(x) = \frac{f(x)}{\tilde{f}(x)}.$$

Then:

$$I = \int_E k(x) \frac{f(x)}{\tilde{f}(x)} \tilde{f}(x)\, dx = \mathbb{E}_{\tilde{f}}\left[k(\tilde{X}) \cdot L(\tilde{X})\right],$$

where $\tilde{X} \sim \tilde{f}$ (tilde-X is sampled from the proposal density).

5.1.7 Importance Sampling – The Change-of-Measure Idea III

Importance Sampling Estimator

Generate $\tilde{X}_1, \ldots, \tilde{X}_R \overset{\text{iid}}{\sim} \tilde{f}$ and compute:

$$\hat{Y}_R^{\text{IS}} = \frac{1}{R} \sum_{j=1}^{R} k(\tilde{X}_j)\, \frac{f(\tilde{X}_j)}{\tilde{f}(\tilde{X}_j)}.$$

Each sample is *weighted* by the likelihood ratio $L(\tilde{X}_j) = f(\tilde{X}_j)/\tilde{f}(\tilde{X}_j)$ to correct for the fact that we sampled from $\tilde{f}$ rather than f.

5.1.7 Importance Sampling – The Change-of-Measure Idea IV

Variance of IS Estimator (Proposition 5.1.33)

The IS estimator is unbiased and strongly consistent. By the CLT:

$$\sqrt{R}(\hat{Y}_R^{\mathrm{IS}} - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2),$$

where:

$$\varsigma^2 = \int_E \left(k(x) \frac{f(x)}{\tilde{f}(x)} \right)^2 \tilde{f}(x)\, dx - I^2 = \mathrm{Var}_{\tilde{f}}\left[k(\tilde{X}) L(\tilde{X}) \right].$$

To reduce variance, choose $\tilde{f}$ so that the weighted integrand $k(x)f(x)/\tilde{f}(x)$ is as *flat* (constant) as possible.

5.1.7 Importance Sampling – The Change-of-Measure Idea V

Optimal Proposal Distribution (Zero-Variance IS)

The variance ς^2 is minimised at $\varsigma^2 = 0$ by the *zero-variance proposal*:

$$\tilde{f}^*(x) = \frac{|k(x)|f(x)}{\int_E |k(x)|f(x)\,dx} = \frac{|k(x)|f(x)}{I'}$$

(proportional to the absolute integrand), because then $k(x)f(x)/\tilde{f}^*(x) = I'$ is constant.
Caveat: This requires knowing the integral I' (essentially knowing I), so it cannot be used directly. However, it guides the choice of $\tilde{f}$: the proposal should be large where $|k(x)|f(x)$ is large, i.e., concentrate samples in the most important regions.

Example 5.1.34 – IS for a Cauchy Tail Probability

Estimate $I = \mathbb{P}(X > 2)$ where X has a Cauchy distribution: $f(x) = 1/(\pi(1 + x^2))$. The true value is $I = 1/2 - \arctan(2)/\pi \approx 0.14758$.

- **CMC:** $\mathrm{Var}(\hat{Y}_R^{\mathrm{CMC}}) = I(1 - I)/R = 0.1258/R$.
- **IS** with proposal $\tilde{f}(x) = \frac{2}{x^2}\mathbf{1}(x > 2)$ (a Pareto-type density concentrated on $(2, \infty)$): Simulating $\tilde{X} \sim \tilde{f}$ via the inverse CDF: $\tilde{X} = 2/U$ where $U \sim \mathcal{U}[0, 1]$.
- IS variance: $\mathrm{Var}(\hat{Y}_R^{\mathrm{IS}}) = 9.55 \times 10^{-5}/R$.
- Reduction: $0.1258/(9.55 \times 10^{-5}) \approx 1317\times$.

5.1.7 IS for Rare Events – Normal Tail I

Example 5.1.36 – IS for $\mathbb{P}(Z > 4)$, $Z \sim \mathcal{N}(0,1)$

The standard normal tail probability $I = \mathbb{P}(Z > 4) \approx 3.167 \times 10^{-5}$ is very small (rare event). CMC is almost useless: most samples give $Y = 0$.

Idea: Shift the proposal to $\tilde{f} = \mathcal{N}(\theta, 1)$ (normal with mean θ, shifted towards the rare region). The likelihood ratio is:

$$L(z) = \frac{\phi_{0,1}(z)}{\phi_{\theta,1}(z)} = e^{-\theta z + \theta^2/2},$$

where $\phi_{0,1}$ is the standard normal density (phi sub 0-1) and $\phi_{\theta,1}$ is the density of $\mathcal{N}(\theta, 1)$. The IS estimator becomes:

$$\hat{Y}_R^{\mathrm{IS},\theta} = \frac{1}{R} \sum_{i=1}^{R} \mathbf{1}(\tilde{Z}_i > 4)\, e^{-\theta \tilde{Z}_i + \theta^2/2}, \quad \tilde{Z}_i \sim \mathcal{N}(\theta, 1).$$

5.1.7 IS for Rare Events – Normal Tail II

Numerical comparison for $I = \mathbb{P}(Z > 4)$:

Proposal θ	Estimate ($R = 10^5$)	Var$/R$	Reduction
0 (CMC)	varies wildly	3.17×10^{-5}	$1\times$
$\theta = 2$	$\approx 3.2 \times 10^{-5}$	5.6×10^{-8}	$566\times$
$\theta = 4$	$\approx 3.2 \times 10^{-5}$	3.6×10^{-8}	$879\times$
$\theta = 4.2256$ (CE opt.)	$\approx 3.2 \times 10^{-5}$	4.5×10^{-9}	$7000\times$

The choice $\theta = 4$ (shifting the proposal mean to the rare-event boundary) gives nearly $900\times$ improvement. The cross-entropy optimal $\theta^* = 4.2256$ does even better (Section 5.1.8).

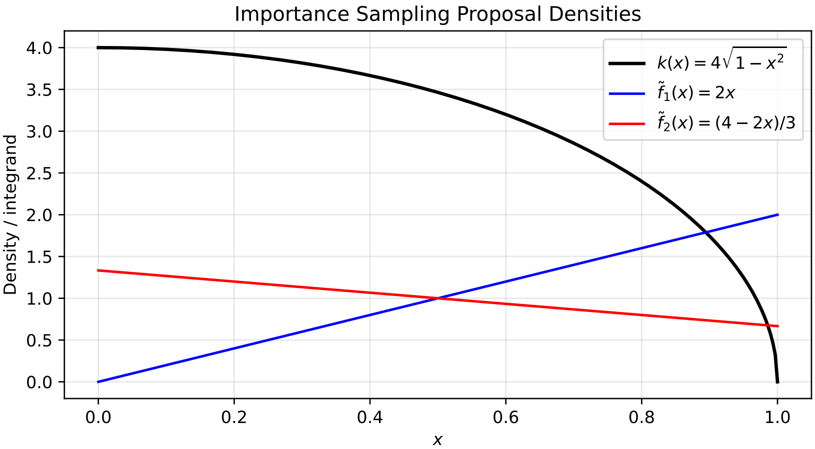
How to Read This

IS for rare events works by moving the entire proposal distribution to where the rare event lives. Samples are then drawn mostly from the region $Z > 4$, and each is downweighted by $e^{-\theta Z + \theta^2/2}$ to correct for the fact that we over-sampled that region.

5.1.7 Importance Sampling – Python Code

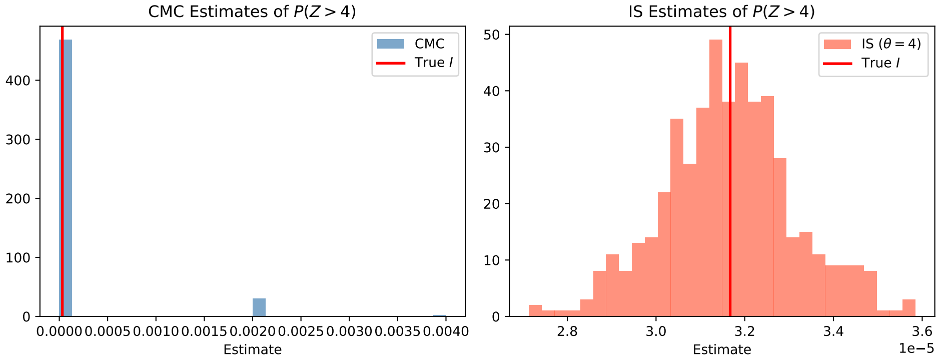
```
import numpy as np
from scipy.stats import norm

def IS_normal_tail(theta, R=100000):
    """Estimate P(Z>4), Z~N(0,1), using IS with N(theta,1) proposal."""
    Z = np.random.normal(theta, 1, R)
    # Log likelihood ratio: log[phi(z;0,1)/phi(z;theta,1)]
    log_LR = -theta * Z + theta**2 / 2
    Y_IS   = (Z > 4).astype(float) * np.exp(log_LR)
    est    = np.mean(Y_IS)
    var    = np.var(Y_IS, ddof=1) / R
    return est, var

np.random.seed(42)
true_I = 1 - norm.cdf(4)
print(f"True P(Z>4) = {true_I:.4e}")
print()
for theta in [0, 2, 4, 4.2256]:
    est, var = IS_normal_tail(theta)
    label = "(CMC)" if theta == 0 else ""
    reduction = true_I * (1 - true_I) / (R := 100000) / var if var > 0 else float('inf')
    print(f"theta={theta:.4f} {label:6s}: est={est:.3e}, "
          f"var/R={var:.3e}, reduction={true_I*(1-true_I)/var:.0f}x")
```

5.1.7 IS Proposal Distributions



The optimal proposal $\tilde{f}^*(x) \propto |k(x)|f(x)$ should closely match the shape of the absolute integrand. Among competing proposals, the one whose shape best matches $k(x)f(x)$ achieves lower variance. A proposal that is too light in the tails inflates variance; too heavy causes high-variance weights.

5.1.7 IS for Rare Events: $\mathbb{P}(Z > 4)$



Top: CMC estimates are mostly exactly 0 (the event $Z > 4$ almost never occurs in random standard normal draws). Bottom: IS with $\tilde{f} = \mathcal{N}(4, 1)$ concentrates all samples in the relevant region $Z > 4$, giving consistent and low-variance estimates.

5.1.7 Self-Normalised Importance Sampling I

When the normalisation constants of both the target f and the proposal $\tilde{f}$ are unknown (a common situation in Bayesian statistics), we can still form a consistent estimator using a ratio (Rao-Blackwell) approach.

Self-Normalised IS Estimator

Suppose $f(x) = c \cdot f_0(x)$ and $\tilde{f}(x) = \tilde{c} \cdot \tilde{f}_0(x)$ with unknown constants. Define unnormalised weights $w(x) = f_0(x)/\tilde{f}_0(x)$. Then:

$$I = \frac{\mathbb{E}_{\tilde{f}}[k(\tilde{X})w(\tilde{X})]}{\mathbb{E}_{\tilde{f}}[w(\tilde{X})]},$$

estimated by the ratio:

$$\hat{Y}_R^{\mathrm{SN}} = \frac{\sum_{i=1}^{R} k(\tilde{X}_i)w(\tilde{X}_i)}{\sum_{i=1}^{R} w(\tilde{X}_i)} = \sum_{i=1}^{R} k(\tilde{X}_i)\, w_i^{\mathrm{norm}},$$

where $w_i^{\mathrm{norm}} = w(\tilde{X}_i)/\sum_j w(\tilde{X}_j)$ are the *normalised weights* (each between 0 and 1, summing to 1). This estimator is *biased* but consistent (bias $\to 0$ as $R \to \infty$).

5.1.7 Self-Normalised Importance Sampling II

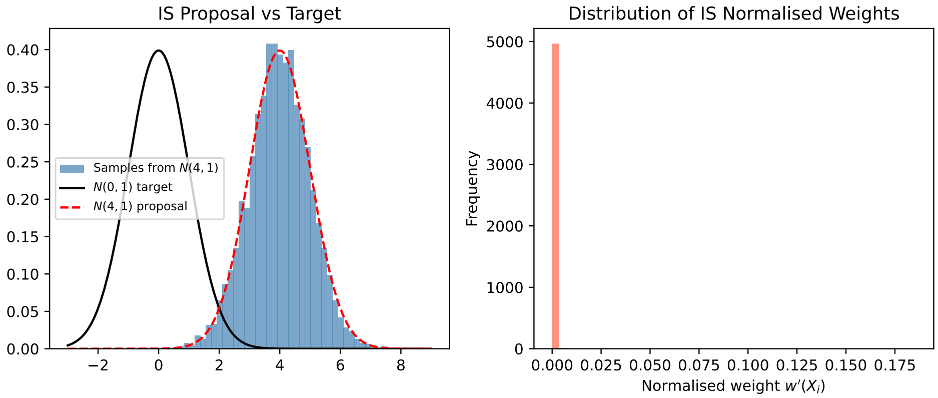
Exponential Change of Measure (Rare-Event Efficiency)

For sum $S_n = X_1 + \cdots + X_n$ of i.i.d. random variables with mean μ (mu, the expected value) and cumulant generating function $\kappa(\theta) = \log \mathbb{E}[e^{\theta X}]$ (kappa of theta): Define the *exponentially twisted* density $f_\theta(x) = e^{\theta x - \kappa(\theta)} f(x)$. Under $\mathbb{P}_\theta$, we have $\mathbb{E}_\theta[X_1] = \kappa'(\theta)$. Choose $\theta = \theta_0$ such that $\kappa'(\theta_0) = \mu + \varepsilon$ (i.e., the twisted mean equals the rare-event threshold). The IS estimator for $I(n) = \mathbb{P}(S_n > n(\mu + \varepsilon))$ becomes:

$$\hat{Y}_R^{\theta_0} = \frac{1}{R} \sum_{i=1}^{R} \mathbf{1}_{S_n^{(i)} > n(\mu+\varepsilon)} \, e^{-\theta_0 S_n^{(i)} + n\kappa(\theta_0)},$$

and this estimator is *logarithmically efficient* (the relative error does not explode as the event becomes rarer).

5.1.7 Self-Normalised IS Weights



Using a proposal $\mathcal{N}(4, 1)$ concentrates samples in the rare region $Z > 4$. The normalised weights $w_i^{\text{norm}} = w(\tilde{X}_i)/\sum_j w(\tilde{X}_j)$ are well-spread (no single sample dominates), ensuring a stable estimate. Poor proposals lead to degenerate weight distributions where one or two samples carry almost all the weight.

5.1.8 The Cross-Entropy Method – Motivation I

The cross-entropy (CE) method provides a systematic procedure to find the *optimal IS proposal* from a parametric family $\{\tilde{f}_\theta : \theta \in \Theta\}$. Rather than choosing θ by hand, CE adapts θ iteratively towards the parameter that minimises the KL divergence from the ideal (zero-variance) proposal.

KL Divergence and Optimal Parameter

The *Kullback-Leibler (KL) divergence* (also called cross-entropy loss) from the zero-variance proposal $g^*(x) = k(x)f(x)/I$ to $\tilde{f}_\theta$ is:

$$D_{\mathrm{KL}}(g^* \| \tilde{f}_\theta) = \int g^*(x) \log \frac{g^*(x)}{\tilde{f}_\theta(x)}\, dx.$$

Minimising this over θ (theta, the parameter vector of the proposal family) is equivalent to maximising:

$$\theta^* = \arg\max_\theta \mathbb{E}_f \left[k(X) \log \tilde{f}_\theta(X) \right].$$

This can be estimated from M samples $(X_1, \ldots, X_M)$ from f as:

$$\hat{\theta}^* = \arg\max_\theta \frac{1}{M} \sum_{i=1}^{M} k(X_i) \log \tilde{f}_\theta(X_i).$$

5.1.8 The Cross-Entropy Method – Motivation III

CE for Rare Event Estimation: Iterative Algorithm

For $I = \mathbb{P}_f(h(\mathbf{X}) \geq \gamma)$ (probability that performance h exceeds threshold γ, where h (h) is the performance function and γ (gamma) is the target level):
Initialise: θ'_0 = initial parameter (e.g., mean of f); ρ (rho) $\in (0,1)$ is the quantile level (e.g., $\rho = 0.05$); α (alpha) $\in (0,1)$ is a smoothing factor.
Iterate for $t = 1, 2, \ldots$:

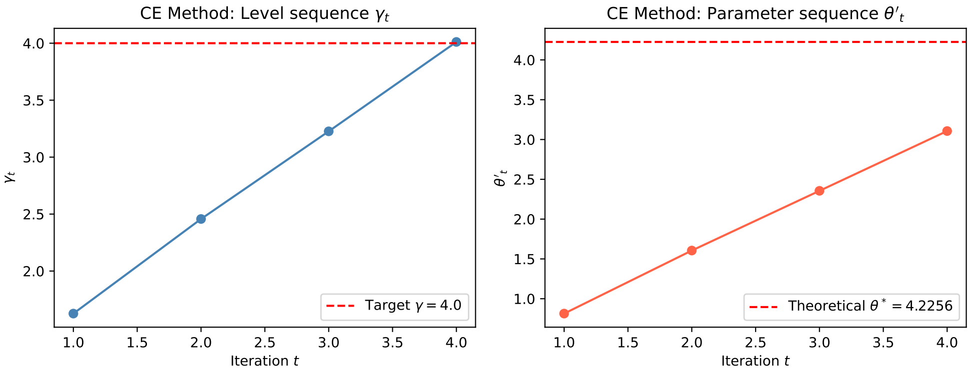
1. **Sample:** Draw $X_1, \ldots, X_M \overset{\text{iid}}{\sim} \tilde{f}_{\theta'_{t-1}}$.
2. **Update level:** $\gamma_t = (1-\rho)$-quantile of $h(X_1), \ldots, h(X_M)$.
3. **Update parameter:**
$$\theta'_t = \arg\max_\theta \frac{1}{M} \sum_{i=1}^{M} \mathbf{1}(h(X_i) \geq \gamma_t)\, L(X_i; f, \tilde{f}_{\theta'_{t-1}})\, \log \tilde{f}_\theta(X_i).$$
4. **Smooth:** $\theta'_t \leftarrow \alpha\theta'_t + (1-\alpha)\theta'_{t-1}$.
5. **Stop** when $\gamma_t \geq \gamma$.

Final IS estimate: Generate R samples from $\tilde{f}_{\theta'_T}$ and compute $\hat{Y}_R^{\text{CE}} = \frac{1}{R}\sum_i \mathbf{1}(h(X_i) > \gamma)\, L(X_i)$.

5.1.8 Cross-Entropy Method – Python Code

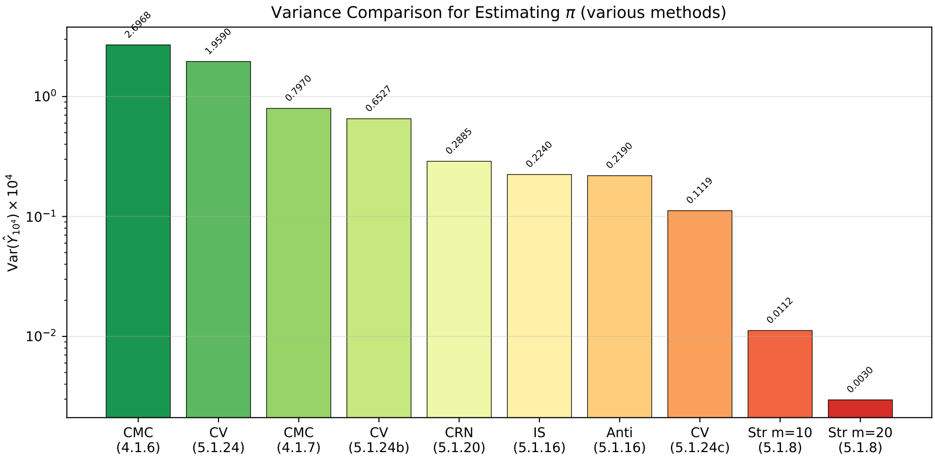
```
import numpy as np
from scipy.stats import norm

def CE_normal_1D(gamma=4.0, M=5000, rho=0.1, alpha=0.7, max_iter=100):
    """CE method to find optimal IS mean for P(Z > gamma), Z~N(0,1)."""
    mu = 0.0
    history = []
    for t in range(max_iter):
        X = np.random.normal(mu, 1, M)
        # Likelihood ratio weights: phi(x;0,1) / phi(x;mu,1)
        w = np.exp(-mu * X + mu**2 / 2)
        # Current level: (1-rho)-quantile
        gamma_t = np.quantile(X, 1 - rho)
        # Update mu: weighted mean of X >= gamma_t
        mask = X >= gamma_t
        if mask.sum() > 0:
            new_mu = np.sum(w[mask] * X[mask]) / np.sum(w[mask])
            mu = alpha * new_mu + (1 - alpha) * mu  # smoothing
        history.append((t+1, gamma_t, mu))
        if gamma_t >= gamma:
            break
    return mu, history

mu_star, hist = CE_normal_1D()
print(f"CE optimal mu* = {mu_star:.4f}  (theory: 4.2256)")
print(f"Converged in {len(hist)} iterations")
# Final IS estimate
```

5.1.8 Cross-Entropy Convergence



Left: The level sequence γ_t (gamma sub t) climbs from 0 to the target $\gamma = 4$ over approximately 46 iterations, with each step adding a fraction $\rho = 10\%$ of the remaining gap. Right: The parameter μ'_t converges to the theoretical CE optimum $\mu^* = 4.2256$. The smoothing factor $\alpha = 0.7$ prevents oscillation.

5.1.9 Summary: Variance Comparison for Estimating π



All methods estimate π using R replications. The variance (plotted on a log scale) shows the improvement of each method over crude CMC.

5.1.9 Summary Table – All Methods

Method	Estimator description	$\mathrm{Var}(\hat{Y}_{10^4})\times 10^4$	Reduction
CMC	$Y = 4\mathbf{1}(U_1^2+U_2^2 \le 1)$	2.697	$1\times$
Cond	$Y = 4\sqrt{1-U^2}$	0.797	$3.4\times$
CV1	$X = \mathbf{1}(U_1+U_2 \le 1)$	1.959	$1.4\times$
CV2	$X = \mathbf{1}(U_1+U_2 \ge \sqrt{2})$	0.653	$4.1\times$
Anti	$Y_1 = 4\sqrt{1-U^2}$, $Y_2 = 4\sqrt{1-(1-U)^2}$	0.219	$12\times$
CRN	$Y_1 = 8/(1+U^2) - 4\sqrt{1-U^2}$	0.289	$9.3\times$
IS	$\tilde{f}(x) = (4-2x)/3$	0.224	$12\times$
CV3	polynomial X, degree 4	0.112	$24\times$
Str	$m = 10$ strata, optimal alloc.	0.0112	$240\times$
Str	$m = 20$ strata, optimal alloc.	0.00296	$911\times$

Key takeaway: methods that exploit the *shape* of the integrand (polynomial CV, optimal stratification) achieve the greatest reductions. Antithetic variates, CRN, and basic IS achieve moderate but practical gains.

5.2 Finance and Actuarial Science – Overview I

Classical financial mathematics models stock prices as *geometric Brownian motion* (GBM). Option pricing requires computing expectations of functionals of GBM paths, which are generally not available in closed form (except for the simple European call option via the Black-Scholes formula). Monte Carlo is the standard tool for complex options, and variance reduction techniques can dramatically reduce computational cost.

Topics Covered in Section 5.2

1. Brownian motion: definition, properties, simulation algorithms
2. Financial contracts: European/Asian options, Black-Scholes formula
3. Stratified sampling for option pricing
4. Control variates for options (geometric mean as CV)
5. Antithetic variates for options
6. Importance sampling for European call options

5.2 Finance and Actuarial Science – Overview II

Running parameters (used in all examples unless noted): $r = 0.05$ (risk-free rate), $\sigma = 0.25$ (sigma, volatility), $S(0) = 100$ (initial stock price), $K = 100$ (strike price).

5.2.1 Brownian Motion – Definition and Properties I

Brownian motion (also called a Wiener process) is the fundamental building block of continuous-time stochastic processes in finance. It models the random fluctuations of a physical particle or financial asset.

Definition 5.2.1 – Standard Brownian Motion

A stochastic process $(B(t) : t \in [0, T])$ is a *standard Brownian motion* if:

1. (i) $B(0) = 0$ (starts at zero).
2. (ii) **Independent increments:** For $0 \leq t_1 < t_2 < \cdots < t_n \leq T$, the increments $B(t_2) - B(t_1), B(t_3) - B(t_2), \ldots$ are independent.
3. (iii) **Normally distributed increments:** $B(t_2) - B(t_1) \sim \mathcal{N}(0, t_2 - t_1)$ for $0 \leq t_1 \leq t_2$. The increment over a time interval of length Δt has standard deviation $\sqrt{\Delta t}$.
4. (iv) $t \mapsto B(t)$ is continuous (no jumps).

Characterisation (Proposition 5.2.2)

A process with continuous paths is a standard BM if and only if:

$$\mathrm{Cov}(B(s), B(t)) = \min(s, t), \quad \text{for all } 0 \leq s \leq t \leq T.$$

This means: the correlation between $B(s)$ and $B(t)$ is $\mathrm{Corr}(B(s), B(t)) = \sqrt{s/t}$ (always positive, increases as $s \to t$).

Simulation on a Grid (Algorithm 36)

On a uniform grid $t_i = i\Delta$, $\Delta = 1/N$, $i = 0, 1, \ldots, N$: generate $Z_1, \ldots, Z_N \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and set:

$$B(i\Delta) = B((i-1)\Delta) + \sqrt{\Delta}\, Z_i.$$

Each increment $\sqrt{\Delta}\, Z_i$ has the correct distribution $\mathcal{N}(0, \Delta)$. In vector form: $(B(1/N), B(2/N), \ldots, B(1)) \overset{d}{=} \frac{1}{\sqrt{N}}(Z_1, Z_1 + Z_2, \ldots, Z_1 + \cdots + Z_N)$.

Brownian Motion with Drift and Geometric Brownian Motion

BM with drift: $X(t) = \sigma B(t) + \mu t$, where μ (mu) is the drift (long-run growth rate) and σ (sigma) is the diffusion coefficient (volatility).
Geometric Brownian Motion (GBM): Under the risk-neutral pricing measure:

$$S(t) = S(0)\exp(\mu^* t + \sigma B(t)), \quad \mu^* = r - \frac{\sigma^2}{2},$$

where r (r) is the risk-free interest rate and $\mu^* = r - \sigma^2/2$ (mu-star) is the risk-neutral drift (adjusted for the convexity of the exponential). For our parameters: $\mu^* = 0.05 - 0.0625/2 = -0.01875$.

5.2.1 Brownian Motion – Definition and Properties V

Brownian Bridge (Bisection Algorithm 38)

The Brownian bridge conditions BM on its endpoint $B(1) = z$. Given $B(0) = 0$ and $B(1) = z$, the midpoint is:

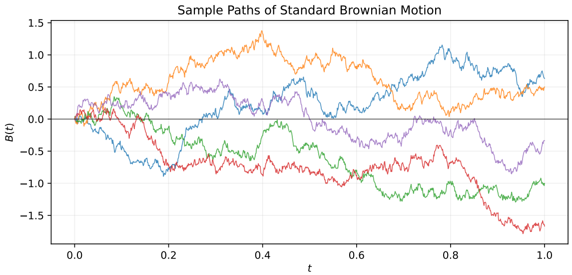
$$(B(1/2) \mid B(1) = z) \sim \mathcal{N}(z/2,\ 1/4).$$

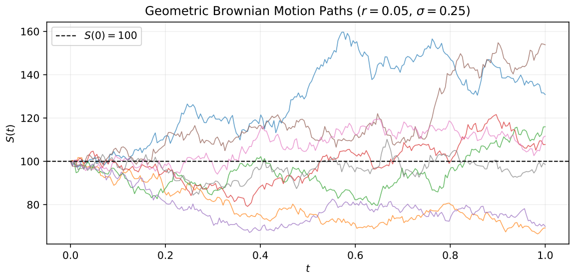
This leads to the bisection algorithm that generates BM in a different order: $B(1), B(1/2), B(1/4), B(3/4), \ldots$ This order is useful for stratified sampling of BM paths.

5.2.1 Brownian Motion – Python Code

```
import numpy as np

def sim_brownian(N=1000, T=1.0, n_paths=5):
    """Simulate standard Brownian motion on [0,T] with N time steps."""
    dt = T / N
    t  = np.linspace(0, T, N + 1)
    # Each column is a BM path; increments are sqrt(dt)*Z
    Z      = np.random.standard_normal((n_paths, N))
    B      = np.zeros((n_paths, N + 1))
    B[:, 1:] = np.cumsum(np.sqrt(dt) * Z, axis=1)
    return t, B

def sim_gbm(S0=100, r=0.05, sigma=0.25, N=252, n_paths=5):
    """Simulate Geometric Brownian Motion (risk-neutral measure)."""
    dt     = 1.0 / N
    mu_star = r - sigma**2 / 2   # risk-neutral drift
    Z      = np.random.standard_normal((n_paths, N))
    log_S = np.zeros((n_paths, N + 1))
    log_S[:, 0] = np.log(S0)
    log_S[:, 1:] = np.log(S0) + np.cumsum(
        mu_star * dt + sigma * np.sqrt(dt) * Z, axis=1)
    return np.linspace(0, 1, N + 1), np.exp(log_S)

np.random.seed(42)
t_bm, paths_bm = sim_brownian(n_paths=3)
t_gbm, paths_gbm = sim_gbm(n_paths=3)
print("BM final values:", paths_bm[:, -1])
```

5.2.1 Brownian Motion Sample Paths





Standard BM $B(t)$: continuous, no drift.
$\mathrm{Var}(B(t)) = t$ grows linearly in time.

GBM $S(t) = S(0)\exp(\mu^* t + \sigma B(t))$:
always positive, log-normally distributed.

5.2.2 Financial Contracts and the Black-Scholes Formula I

A financial *option* gives the holder the right (but not obligation) to buy or sell an asset at a specified price. Under the risk-neutral pricing framework, the option price equals the discounted expected payoff.

Common Option Types Under GBM

All options are priced as $I = e^{-r}\,\mathbb{E}[f(\text{path})]$ where e^{-r} (e to the minus r) is the discount factor for a 1-year option.

- **European call:** $f = (S(1) - K)_+ = \max(S(1) - K, 0)$. Pays the excess of the final stock price $S(1)$ over strike K (capital K).
- **European put:** $f = (K - S(1))_+$.
- **Asian call:** $f = (A - K)_+$ where $A = \frac{1}{n}\sum_{j=1}^{n} S(j/n)$ is the *arithmetic average* price over n monitoring dates.
- **Up-and-out call:** $(S(1) - K)_+ \cdot \mathbf{1}(\max_t S(t) < b)$ (European call that is cancelled if price hits a barrier b).
- **Basket call:** $(\sum_j w_j S_j(1) - K)_+$ for a portfolio of d stocks.

5.2.2 Financial Contracts and the Black-Scholes Formula II

Black-Scholes Formula (European Call, Theorem 5.2.7)

For a European call with $S(t) = S(0)\exp(\mu^* t + \sigma B(t))$:

$$I = e^{-r}\,\mathbb{E}[(S(1) - K)_+] = S(0)\Phi(d_1) - Ke^{-r}\Phi(d_2),$$

where Φ (Phi, capital phi) is the standard normal CDF,

$$d_1 = \frac{1}{\sigma}\left[\ln\left(\frac{S(0)}{K}\right) + r + \frac{\sigma^2}{2}\right], \quad d_2 = d_1 - \sigma.$$

Here d_1 and d_2 are standardised log-moneyness measures: $d_1 > 0$ means the option is currently profitable (in the money).

Worked Example: Black-Scholes Price

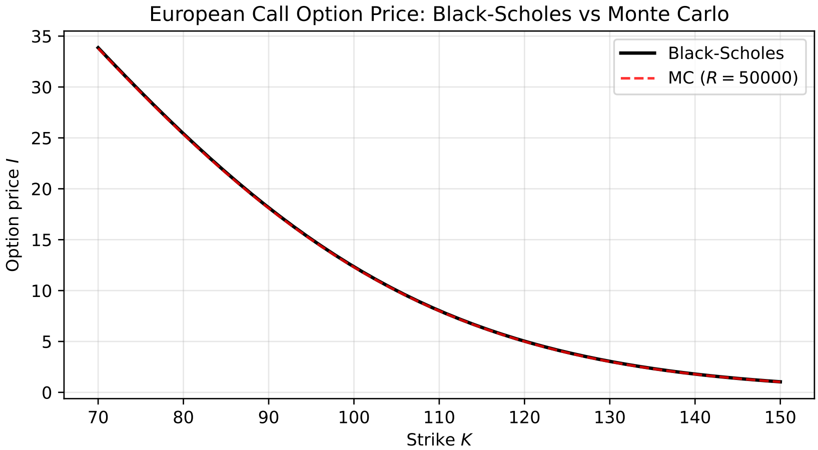
With $S(0) = K = 100$, $r = 0.05$, $\sigma = 0.25$:

$$d_1 = \frac{1}{0.25}\left[\ln(1) + 0.05 + 0.0625/2\right] = \frac{0.08125}{0.25} = 0.325, \quad d_2 = 0.325 - 0.25 = 0.075.$$

$$I = 100\,\Phi(0.325) - 100\,e^{-0.05}\,\Phi(0.075) = 100(0.6274) - 100(0.9512)(0.5299) \approx 12.336.$$

This is the *exact* European call price; Monte Carlo approximates it.

5.2.2 Black-Scholes vs. Monte Carlo



The Black-Scholes formula gives the exact European call price as a function of strike K. Monte Carlo approximates it with controlled statistical error. For the Asian option (no closed form), Monte Carlo is the primary tool, making variance reduction essential.

5.2.3 Stratified Sampling for Option Pricing I

Example 5.2.5 – European Call with Stratified $B(1)$

The European call depends only on $S(1) = S(0)\exp(\mu^* + \sigma B(1))$, and $B(1) \sim \mathcal{N}(0,1)$. Stratify on $Z = B(1)$ using m equally probable strata:

$$A^j = \left(\Phi^{-1}\left(\frac{j-1}{m}\right), \Phi^{-1}\left(\frac{j}{m}\right)\right], \quad j = 1, \ldots, m,$$

where Φ^{-1} (Phi inverse) is the quantile function of $\mathcal{N}(0,1)$. Each stratum has probability $p_j = 1/m$. Within stratum j, sample $V^j = (j-1)/m + U/m$, $U \sim \mathcal{U}[0,1]$, and set $Z^j = \Phi^{-1}(V^j)$.

Results from Table 5.13 ($R = 10^5$, $S(0) = K = 100$, $r = 0.05$, $\sigma = 0.25$):

Strata m	CMC	5	50	200
Estimate	12.280	12.351	12.338	12.336
Half-width b	0.115	0.043	0.025	0.002
Var reduction	–	$7.2\times$	$21\times$	$3778\times$

5.2.3 Stratified Sampling for Option Pricing II

With $m = 200$ strata and optimal allocation, variance is reduced by a factor of nearly 4000, meaning we can achieve the same accuracy with $4000\times$ fewer simulations!

5.2.3 Stratified Sampling for Option Pricing III

Example 5.2.8 – Asian Call with Stratified Brownian Motion

For the Asian option with payoff $A = \frac{1}{n}\sum_{j=1}^{n} S(j/n)$, stratification of a single $Z = B(1)$ already helps. Two methods:

1. **Method 1**: Stratify on $Z = B(1) \sim \mathcal{N}(0,1)$. After fixing $B(1)$, the remaining path is a Brownian bridge.
2. **Method 2**: Use the bisection algorithm (Algorithm 38) and stratify on the first two standard normal variates Z_{2^k} and $Z_{2^{k-1}}$ jointly, creating m^2 2D strata.

Results with $K = 100$, $m = 200$: variance reduced by factor $\approx 84\times$.

5.2.4 Control Variates for Asian Option Pricing I

The Asian call option payoff $e^{-r}(A-K)_+$ with arithmetic mean $A = \frac{1}{n}\sum_j S(j/n)$ has no known closed-form price. However, the *geometric mean* $G = (\prod_j S(j/n))^{1/n}$ *does* have a known distribution under GBM, making it an excellent control variate.

5.2.4 Control Variates for Asian Option Pricing II

Control Variate: Geometric Mean (Example 5.2.9)

Under GBM, the geometric mean is:

$$G = S(0)\exp[\mu_N + \sigma_N Z], \quad Z \sim \mathcal{N}(0,1),$$

where $\mu_N = \mu^*(n+1)/(2n)$ and $\sigma_N^2 = \sigma^2(n+1)(2n+1)/(6n^2)$. The expected discounted geometric-mean payoff $W = e^{-r}\mathbb{E}(G-K)_+$ is computable via Black-Scholes (since G is log-normal). The control variate estimator for the arithmetic Asian option:

$$\hat{I}_R^{\mathrm{cv}} = \hat{I}_R^{\mathrm{CMC}} + \hat{c}(\hat{W}_R - W),$$

where $\hat{c}$ (c-hat) is estimated from pilot simulations and W (capital W) is the known exact geometric-mean option price.

5.2.4 Control Variates for Asian Option Pricing III

Results from Table 5.17 ($R = 10^4$, $n = 10$, $r = 0.05$, $\sigma = 0.25$):

K	$\hat{Y}^{\mathrm{CMC}}$	Error$^{\mathrm{CMC}}$	$\hat{Y}^{\mathrm{CV}}$	Error$^{\mathrm{CV}}$	Reduction
150	0.0919	2.447	0.0586	0.374	$\approx 43\times$
120	1.5706	10.54	1.4219	0.586	$\approx 323\times$
100	7.9348	21.50	7.3853	0.731	$\approx 865\times$
50	51.321	30.66	50.219	0.998	$\approx 942\times$

The control variate is most effective for deep out-of-the-money options ($K = 150$, where payoff is rare) and deep in-the-money options ($K = 50$), achieving up to $\approx 940\times$ variance reduction.
The arithmetic mean A and geometric mean G are strongly correlated (both monotonically depend on the same BM path), so ρ^2 is close to 1.

5.2.5 Antithetic Variates for Asian Call Option II

Construction for Asian Call (Example 5.2.10)

Generate $Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ and define two paths:

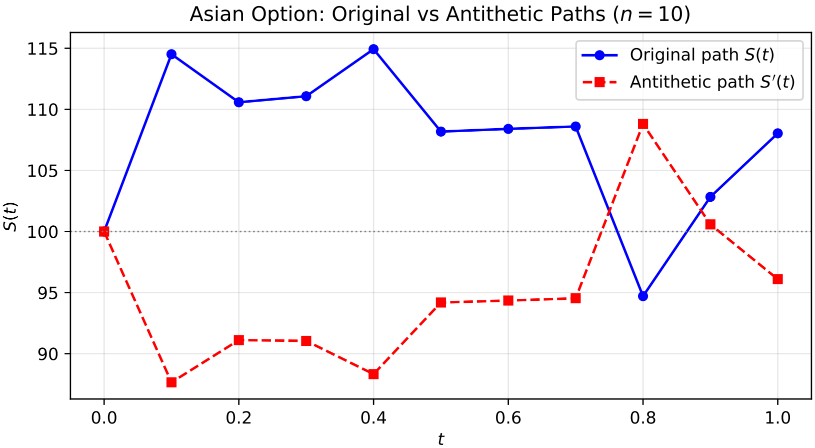
$$B(k/n) = \frac{1}{\sqrt{n}} \sum_{j=1}^{k} Z_j \quad \text{(original)},$$

$$B'(k/n) = \frac{1}{\sqrt{n}} \sum_{j=1}^{k} (-Z_j) \quad \text{(antithetic: flip signs)}.$$

Both paths are valid standard Brownian motions (since $-Z_j \sim \mathcal{N}(0,1)$). The original GBM path:
$S^i(k/n) = S(0)\exp(\mu^*(k/n) + \sigma B(k/n))$. The antithetic path:
$S^{i,\prime}(k/n) = S(0)\exp(\mu^*(k/n) - \sigma B(k/n))$.
The antithetic estimator for the Asian call:

$$\hat{I}_R^{\text{anti}} = \frac{1}{R} \sum_{i=1}^{R} \frac{e^{-r}(A^i - K)_+ + e^{-r}(A^{i,\prime} - K)_+}{2},$$

5.2.5 Antithetic Variates – Path Illustration



The antithetic GBM path $S'(t) = S(0)\exp(\mu^* t - \sigma B(t))$ mirrors the original $S(t) = S(0)\exp(\mu^* t + \sigma B(t))$. When $S(t)$ tends upward (over-estimating the mean), $S'(t)$ tends downward (under-estimating), so their average payoffs cancel the fluctuation.

5.2.6 Importance Sampling for European Call I

IS Estimator for European Call (Example 5.2.11)

The European call price is:

$$I = e^{-r} \int_{-\infty}^{\infty} (S(0)e^{\mu^* + \sigma z} - K)_+ \phi_{0,1}(z)\, dz,$$

where $\phi_{0,1}(z)$ (phi sub 0-1) is the standard normal density evaluated at z. Use proposal $\tilde{f} = \phi_{x^*,1}$ (shifted normal with mean x^*). The IS estimator:

$$\hat{Y}_R^{\mathrm{IS},x^*} = \frac{1}{R} \sum_{i=1}^{R} e^{-r} (S(0)e^{\mu^* + \sigma \tilde{Z}_i} - K)_+ \, e^{-x^* \tilde{Z}_i + (x^*)^2/2},$$

where $\tilde{Z}_i \sim \mathcal{N}(x^*, 1)$ (tilde-Z, sampled from the shifted normal). The term $e^{-x^* \tilde{Z}_i + (x^*)^2/2}$ is the likelihood ratio $\phi_{0,1}(\tilde{Z}_i)/\phi_{x^*,1}(\tilde{Z}_i)$.

Two Methods for Choosing the Shift x^*

1. **Heuristic:** Find the mode of the integrand $h(z) = (S(0)e^{\mu^*+\sigma z} - K)_+ \phi_{0,1}(z)$ by solving $h'(z) = 0$, which gives the equation: $S(0)\sigma e^{\mu^*+\sigma z} = z(S(0)e^{\mu^*+\sigma z} - K)$. For $S(0) = K = 100$, $r = 0.05$, $\sigma = 0.25$: $x^* \approx 1.033$.
2. **Cross-entropy:** Estimate the CE-optimal shift from M pilot samples:

$$\hat{x}^*_{\mathrm{CE}} = \frac{\sum_i Z_i \, (S(0)e^{\mu^*+\sigma Z_i} - K)_+}{\sum_i (S(0)e^{\mu^*+\sigma Z_i} - K)_+},$$

the payoff-weighted mean of Z. For our parameters: $x^*_{\mathrm{CE}} \approx 1.232$.

5.2.6 Importance Sampling for European Call III

Results from Table 5.18 ($S(0) = K = 100$, $r = 0.05$, $\sigma = 0.25$, $R = 10^4$):

	CMC	IS ($x^* = 1.033$)	IS CE ($x^* = 1.232$)
Estimate $\hat{Y}$	12.391	12.302	12.306
Half-width b	0.387	0.125	0.120
Var reduction	–	$9.65\times$	$10.46\times$

Both IS methods achieve approximately $10\times$ variance reduction. The CE method gives a slightly smaller half-width.

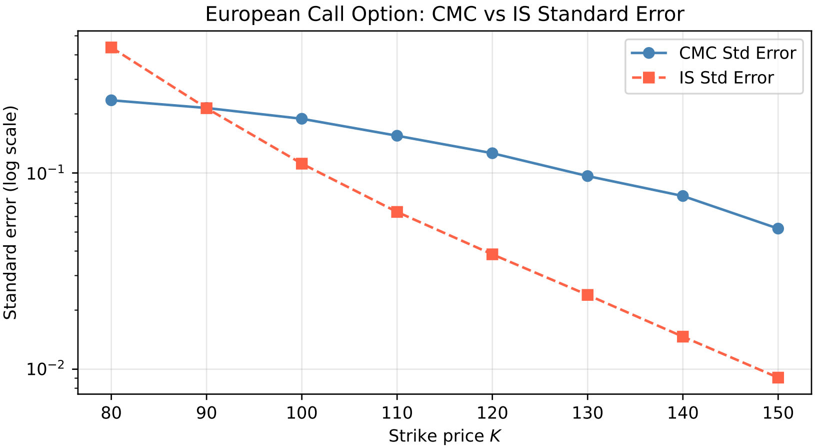
Conclusion

IS is most useful for far out-of-the-money options (large $K/S(0)$), where the payoff region is rare and the optimal shift x^* can be very large. For at-the-money options, stratified sampling and CV tend to outperform IS.

5.2.6 IS for European Call – Python Code

```
import numpy as np
from scipy.stats import norm

def european_call_IS_CE(S0=100, K=100, r=0.05, sigma=0.25, R=10000, M=10000):
    """European call price via IS with CE-optimal shift x*."""
    mu_star = r - sigma**2 / 2
    # CE step: estimate x* as payoff-weighted mean of Z
    Z_pilot = np.random.standard_normal(M)
    payoff   = np.maximum(S0 * np.exp(mu_star + sigma * Z_pilot) - K, 0)
    x_star   = np.sum(Z_pilot * payoff) / np.sum(payoff)
    print(f"  CE x* = {x_star:.4f}")

    # IS estimation
    Z_tilde = np.random.normal(x_star, 1, R)
    S1      = S0 * np.exp(mu_star + sigma * Z_tilde)
    payoff_IS = np.maximum(S1 - K, 0)
    log_LR    = -x_star * Z_tilde + x_star**2 / 2   # log(phi_{0,1}/phi_{x*,1})
    Y_IS      = np.exp(-r) * payoff_IS * np.exp(log_LR)
    est  = np.mean(Y_IS)
    b    = 1.96 * np.std(Y_IS, ddof=1) / np.sqrt(R)
    return est, b

# CMC for comparison
def european_call_CMC(S0=100, K=100, r=0.05, sigma=0.25, R=10000):
    mu_star = r - sigma**2 / 2
    Z = np.random.standard_normal(R)
    Y = np.exp(-r) * np.maximum(S0 * np.exp(mu_star + sigma * Z) - K, 0)
```

5.2.6 IS for European Call – Standard Error Comparison



IS achieves substantially lower standard error for out-of-the-money options (large K/S_0), where CMC rarely generates non-zero payoffs. The shift x^* moves the sampling distribution toward the payoff region, making the rare event common under the proposal.

5.3 Bibliographical Notes

Key references for each topic:

- **Stratified sampling and variance reduction fundamentals:** Rubinstein & Kroese [186]; Asmussen & Glynn [8].
- **Antithetic variates, CRN, control variates:** Asmussen & Glynn [8]; Ross [181].
- **Network reliability estimation:** Madras [138] (the 11-node, 22-edge network example).
- **Cross-entropy method (original paper):** Rubinstein [184] (1999); comprehensive treatment: Rubinstein & Kroese [186].
- **CE tutorial:** de Boer et al. [31].
- **Logarithmic efficiency and exponential change of measure:** Asmussen & Glynn [8].
- **Brownian motion bisection method:** Asmussen & Glynn [8]; Knuth [106].
- **Finance – Black-Scholes:** Black & Scholes (1973); Hull [96].
- **Asian option pricing with IS and CV:** Kemna & Vorst (1990) (geometric mean CV); Glasserman [74].

Exercise 5.T.1 (Stratified Sampling with Unequal Costs)
Suppose we estimate $I = \mathbb{E}[Y]$ with strata $A^1, \dots, A^m$ and $p_j = \mathbb{P}(Y \in A^j)$. The cost per replication in stratum j is $t_j > 0$ (different strata may require different computation times). Total budget C (not R).

- (a) Show that the stratified estimator satisfies a CLT with $\zeta^2 = \sum_{j=1}^m \frac{p_j^2 \sigma_j^2}{\gamma_j} \sum_{k=1}^m t_k \gamma_k$ where $\gamma_j = \lim R_j t_j / C$ is the fraction of budget allocated to stratum j.
- (b) Show that the optimal budget allocation (minimising ζ^2) is $\gamma_j \propto p_j \sigma_j / \sqrt{t_j}$, i.e., assign more budget to strata that are more variable *and* cheaper to simulate.

Exercise 5.T.5 (Antithetic for Exponential)
Let $Y = e^U$, $U \sim \mathcal{U}[0,1]$. The antithetic estimator uses $(e^U + e^{1-U})/2$. Compute $\mathrm{Corr}(e^U, e^{1-U})$ exactly and confirm the $\approx 51\times$ variance reduction claimed in Example 5.1.13. (*Hint:* $\mathbb{E}[e^U e^{1-U}] = e\,\mathbb{E}[1] = e$.)

Exercise 5.T.8 (Optimal IS Distribution)
Let $k : E \to \mathbb{R}_{\geq 0}$ (non-negative). Show that the proposal density $g^*(x) = k(x)f(x)/I$ minimises the IS variance: $\arg\min_g \operatorname{Var}_g(k(X)f(X)/g(X)) = g^*$.
Hint: Jensen's inequality gives $\mathbb{E}_g\big[(k(X)f(X)/g(X))^2\big] \geq \big(\mathbb{E}_g[k(X)f(X)/g(X)]\big)^2 = I^2$, with equality iff $k(x)f(x)/g(x)$ is constant $\mathbb{P}_g$-a.s.

Exercise 5.T.25 (CE Optimal Parameter for Normal Family)
For $I = \mathbb{P}(Z > \gamma)$, $Z \sim \mathcal{N}(0,1)$, show that the CE-optimal mean for the family $\{\mathcal{N}(\theta,1)\}$ is:

$$\theta^* = \frac{\sqrt{2}\, e^{-\gamma^2/2}}{\sqrt{\pi}\big(1 - \operatorname{erf}(\gamma/\sqrt{2})\big)},$$

where $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$ is the error function. For $\gamma = 4$: $\theta^* = 4.2256$.

Lab Exercise 5.L.1
Implement and compare CMC and antithetic variate estimation of $I = \int_0^1 e^x\, dx = e - 1$. Use $R = 10^4$ replications. Verify that the antithetic estimator achieves the theoretical $\approx 51\times$ variance reduction.

Chapter 5 Summary I

Core Message

All variance reduction techniques aim to construct an estimator $\hat{Y}_R^{\text{new}}$ with asymptotic variance $(\varsigma^{\text{new}})^2 < \operatorname{Var}(Y)$, equivalent to obtaining more information per simulation run. The key is to exploit the *structure* of the problem.

Method	**Key idea**	**Best used when**
Stratified sampling	Divide input space into strata; allocate samples optimally	Function is non-constant; strata means differ
Antithetic variates	Generate negatively correlated pairs (Y, Y')	Integrand is monotone in U_i
Common random numbers	Use same stream for two systems	Comparing two system configurations
Control variates	Correct estimate via $c(\hat{X}_R - \mu_X)$	Correlated auxiliary with known mean
Conditional MC	Compute $\mathbb{E}[Y\|X]$ analytically	Conditional expectation is tractable
Importance sam-	Change measure to $\tilde{f}$	Rare events; integrand concen-

Chapter 5 Summary II

Finance summary (Asian call, $R = 10^4$, $n = 10$, $K = 150$):

- CMC: half-width $b = 2.45$
- Control variates (geometric mean): $b = 0.374$ ($\approx 43\times$ reduction)
- Antithetic variates: $b = 0.015$ ($\approx 25{,}000\times$ reduction!)
- Stratified BM: $b \approx 0.03$ for $m = 200$ strata ($\approx 84\times$ reduction)
- IS + CE for European call: $\approx 10\times$ reduction

Final Takeaway

No single method dominates in all situations. Antithetic variates excel for deep out-of-the-money options. Control variates are most effective near at-the-money. Stratified sampling is most powerful for smooth integrands. In practice, methods can be combined (e.g., stratification + antithetics).